

Tutorial 11

Exercise 1: Modelling the knapsack problem

Model the optimization variant of knapsack as a 0-1 ILP.

Exercise 2: Modelling the subset sum problem

Model the decision variant of subset sum as a 0-1 ILP.

Exercise 3: Modelling 3-coloring

Model the decision variant of 3-coloring (see Lecture 6, Slide 8) as a 0-1 ILP.

Exercise 4: ILP in NP

Show that 0-1 ILP is in NP.